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INTRODUCTION

The implementation and study of Armstrong's Indirect Method of FM Generation, a crucial method in contemporary communication systems that generates high-stability wideband frequency modulation (WFM), are examined in this report. Armstrong's approach first creates a Narrow-Band FM (NBFM) signal using a crystal oscillator and a balanced modulator, in contrast to direct approaches that alter the oscillator frequency, often at the expense of carrier stability. The appropriate wideband frequency deviation and carrier frequency are then obtained by processing this NBFM signal through a number of frequency multipliers and mixers.

In this lab, we simulate the Armstrong method using MATLAB/Simulink, centered our FM spectrum at $f_c = 25\text{kHz}$ to see the modulated signal's spectral properties. The practical design of an FM transmitter operating at 100MHz with a frequency variation of 72kHz is also examined in this report. We intend to verify the connection between frequency multiplication and modulation index enhancement by moving from the theoretical framework of NBFM, represented by $\phi_{FM}(t) = A_c \cos(2\pi f_c t) - K_f a(t) \sin(2\pi f_c t)$, to a fully realized two-stage indirect generation design.

PROCEDURE

In order to simulate Armstrong's indirect method, a 1 kHz integrated message signal, a 10 kHz carrier, and its quadrature version are combined to create a Narrow-Band FM (NBFM) signal. This implements the approximation $\phi_{FM}(t) = A_c \cos(2\pi f_c t) - K_f a(t) \sin(2\pi f_c t)$.

After that, a Polyval block set to the fourth-degree polynomial x^4 is applied to this 10 kHz NBFM signal. This block functions as a frequency multiplier, raising the carrier frequency to 40 kHz and the frequency deviation by a factor of four.

The 40 kHz signal is combined with a local oscillator and processed through a bandpass filter to isolate the appropriate difference frequency in order to obtain the final target carrier of 25 kHz. A Spectrum Analyzer is used to record the final output, which offers a thorough view of the FM spectrum properties centered at 25 kHz. This sequence illustrates the fundamental idea of Armstrong's approach, which involves generating NBFM utilizing low-frequency stable oscillators, then achieving the wideband frequency and deviation needed for high-fidelity broadcasting by non-linear multiplication and mixing.

$$\text{Narrow-Band FM (NBFM): } \phi_{FM}(t) \approx A_c \cos(\omega_c t) - K_f a(t) A_c \sin(\omega_c t)$$

Armstrong Simulation

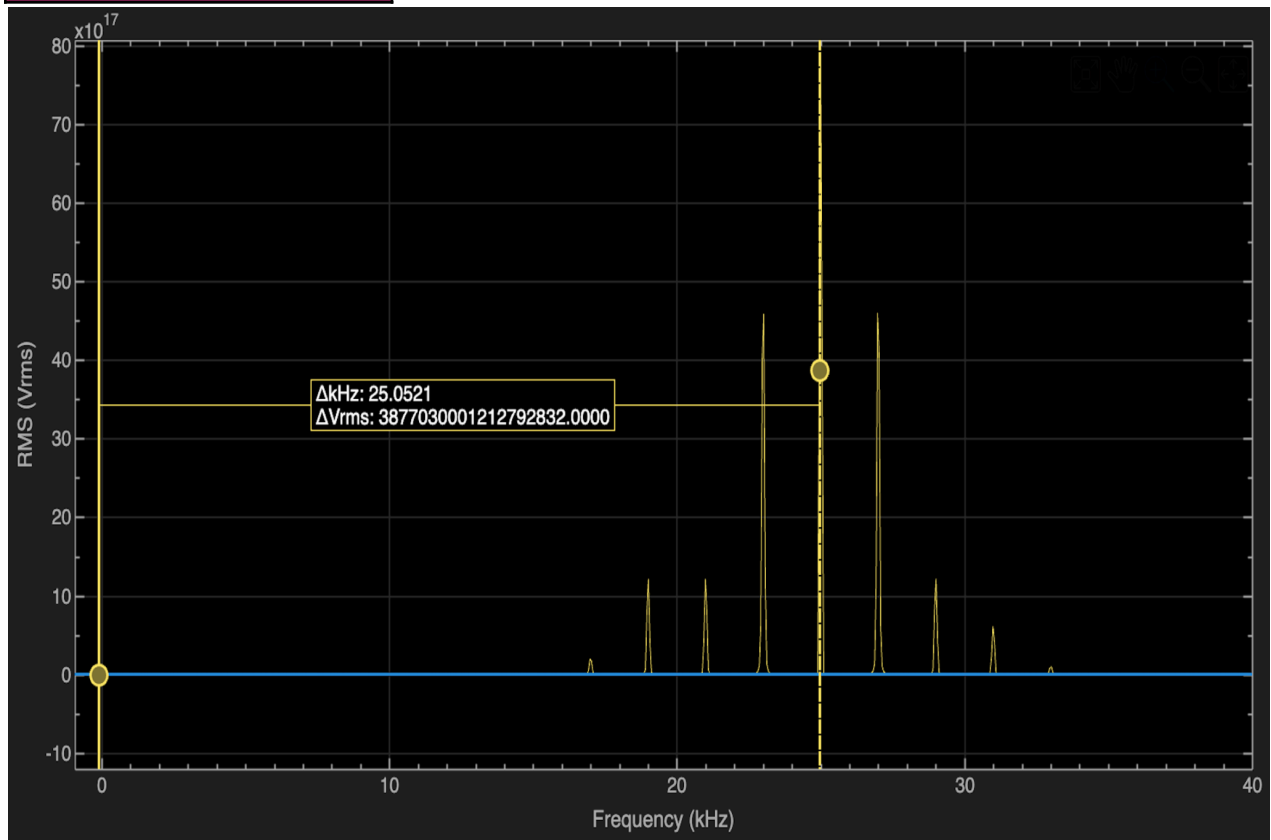


Figure 1 - FM spectrum centred at $f_c = 25$ kHz

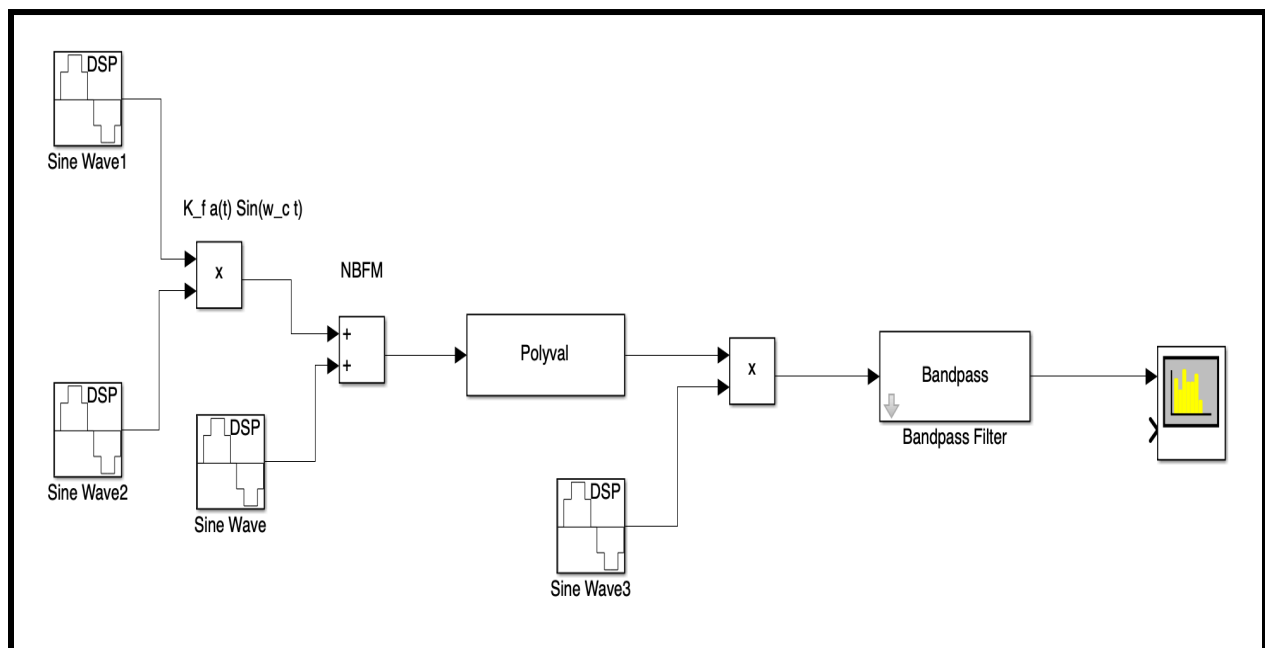


Figure 2 - Simulink block diagram of Armstrong's indirect method

Post Lab

Specifications: Design a circuit that produces FM waves at 100 MHz with a maximum frequency deviation of 72 kHz, A NBFM system that produces FM at 150 kHz with a maximum frequency deviation of 18 Hz is available. A 10 MHz oscillator is also available.

- Draw a block diagram with all the necessary components as the blocks.
 - Indicate the input and output of each block clearly.
 - Clearly label each block to indicate its function.
 - Include any additional circuit blocks you might need.
- **NOTE:** Use Armstrong's indirect method in two stages.

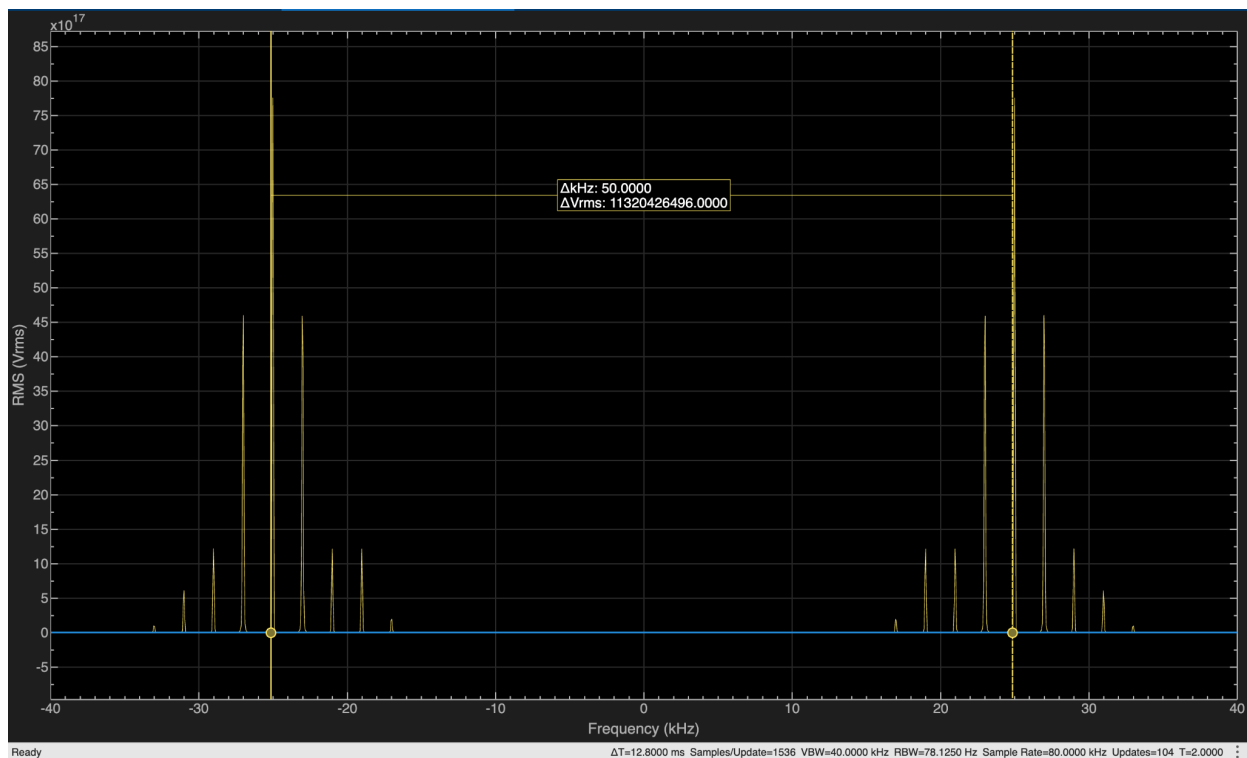


Figure 3 - Narrow-Band Frequency Modulation (NBFM) Signal, $\phi_{FM}(t)$

The spectral output of the Armstrong indirect method simulation is shown in the given graph in Figure 3, which shows a frequency-modulated signal centered at the intended carrier frequency of 25 kHz. After scaling an original 10 kHz NBFM source through a frequency multiplication factor of four and a down-conversion mixing stage, the spectrum displays a symmetric distribution of sidebands around the carrier, which is typical of FM signals. The relocated carrier is shown by the major peaks at $\pm 25\text{kHz}$, and the neighboring impulses show that the x^4 polynomial block has successfully widened the frequency deviation. This graphic demonstrates that the simulation accurately converted the modulated signal to the intended intermediate frequency while maintaining the FM spectral features.

In order to scale the initial 18Hz frequency deviation up to the desired 72kHz, the Armstrong indirect FM transmitter design necessitates a two-stage frequency multiplication process.

$$N_{total} = \frac{\Delta f_{target}}{\Delta f_{initial}} = \frac{72,000}{18} = 4000$$

N_{total} is the total required multiplication factor that we first compute. We divided the multiplication into two stages, N_1 and N_2 , with a frequency mixer in between to down-convert the carrier as a single stage would change the carrier frequency to an uncontrollable range (600 MHz).

The 150kHz NBFM signal is multiplied ($N_1 = 66.66$) in the first stage, raising the carrier to 12MHz and the deviation to 1.44kHz. A mixer utilizes the 10MHz oscillator to translate the carrier down to 2MHz (12MHz – 10MHz) while maintaining the 1.44kHz deviation in order to prevent the second stage from pushing the carrier past the 100MHz target. The second multiplier ($N_2 = 60$) receives this intermediate signal and concurrently scales the 2MHz carrier to the final 100MHz and the 1.44kHz deviation to the necessary 72kHz.

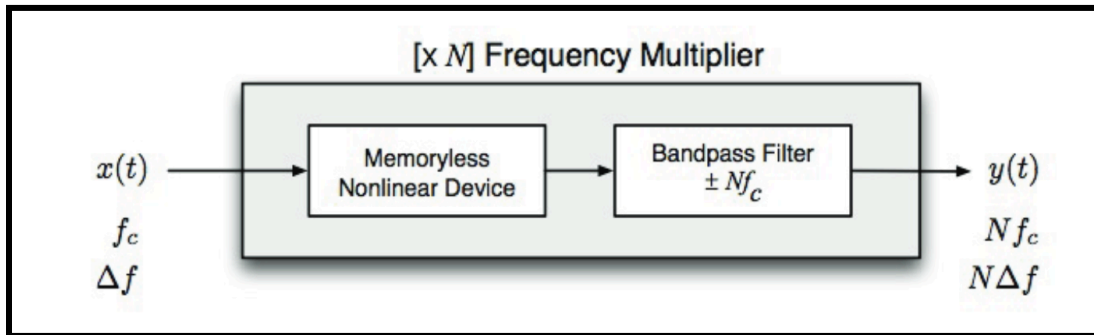


Figure 3 - Block Diagram and Functionality of a Frequency Multiplier

The NBFM source, Frequency Multiplier 1 ($\times 66.66$), a Balanced Mixer connected to a 10MHz Local Oscillator, a Bandpass Filter to choose the difference frequency, and a final Frequency Multiplier 2 ($\times 60$) make up the resulting block diagram. The system may achieve exact broadcast specifications while preserving the frequency stability of the original crystal-controlled NBFM source thanks to its architecture, which guarantees that the carrier frequency and frequency deviation are scaled independently through mixing.

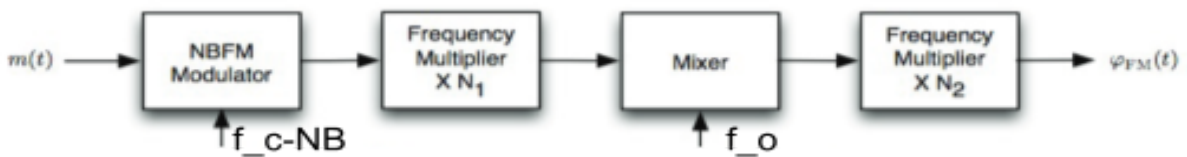


Figure 4 - Block Diagram of Indirect Method of Armstrong for FM Generation in two (2) Stages

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Lab Assignment 4: Armstrong's Indirect Method of FM Generation

Narrow-Band Frequency Modulation (NBFM) Signal :

$$\phi_{FM}(t) \approx A_c \cos(\omega_c t) - K_f a(t) \cdot A_c \sin(\omega_c t)$$

• Initial NBFM : $f_{cNB} = 150 \text{ KHz}$, $\Delta f_{NB} = 18 \text{ Hz}$

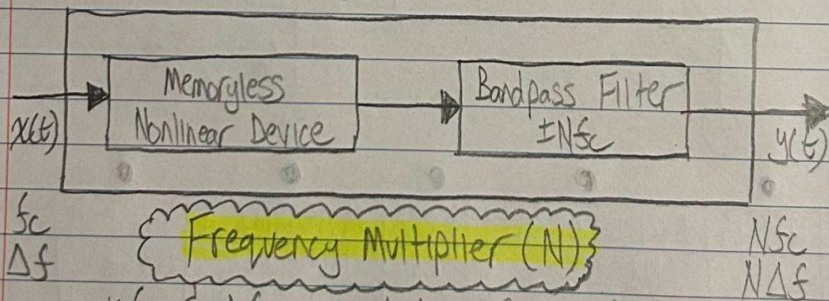
• Target WBFM : $f_{cWB} = 100 \text{ KHz}$, $\Delta f_{WB} = 72 \text{ KHz}$

• Oscillator Frequency : $f_o = 10 \text{ MHz}$

■ Determine the total multiplication factor (N)

$$N = \frac{\Delta f_{WB}}{\Delta f_{NB}} = \frac{72000}{18} = 4000$$

• Since we are using Armstrong's Indirect method to generate a WBFM signal in two (2) stages, the multiplication factor N is split into two stages where $(N_1)(N_2) = 4000$ and a frequency converter (mixer) in between

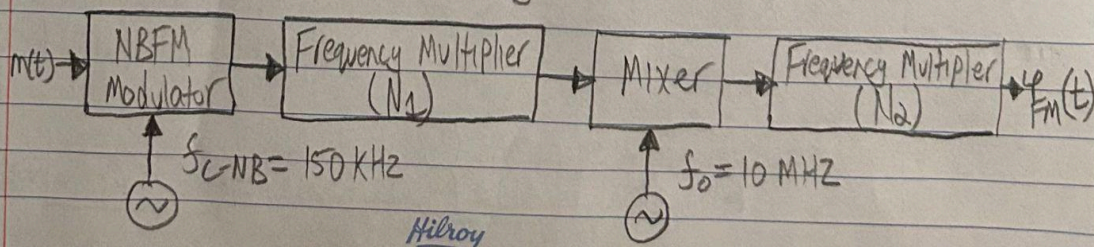


• f_c & Δf increased by a factor of N

• Since $\beta = \frac{\Delta f}{B_m}$, modulation index increased by a factor of N

• This forms basis for $\phi_{NBFM}(t)$ to $\phi_{WBFM}(t)$

■ Indirect Method of Armstrong



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(2)

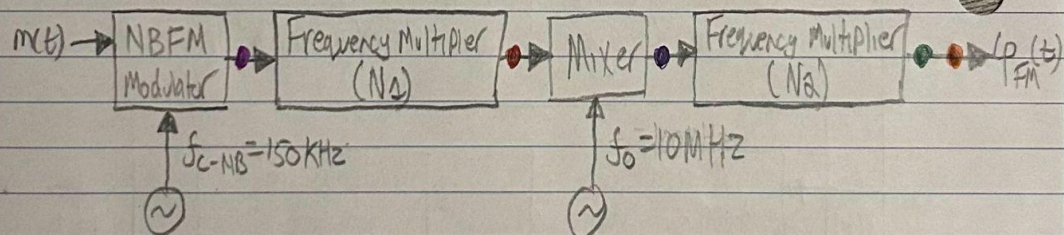
- From the previous block diagram of Armstrong's Indirect method, the first block is the NBFM modulator and holds a condition for the generation of WBFM, the modulation index, $\beta < 0.3$

* Since this circuit generates FM waves at 100 MHz

$$\hookrightarrow \beta = \frac{\Delta f_{NB}}{100 \text{ MHz}} = \frac{18}{100 \text{ MHz}} = 1.8 \times 10^{-7}$$

$\therefore$ the NBFM condition, $\beta < 0.3$, is satisfied

- Now, going back to the two (2) stage circuit for FM signal generation, splitting the multiplication factor (N), such that $(N_1)(N_2) = 4000$. This, we choose $N_1 = 50$ & $N_2 = 80$



	NBFM	Multiplier 1	Mixer	Multiplier 2	WBFM
Δf	Δf_{NB}	$N_1 \cdot \Delta f_{NB}$	$N_1 \cdot \Delta f_{NB}$	$N_2 N_1 \Delta f_{NB}$	Δf_{WB}
f_c	f_{c-NB}	$N_1 \cdot f_{c-NB}$	$f_o - N_1 \cdot f_{c-NB}$	$N_2 \cdot f_o - N_2 N_1 \cdot f_{c-NB}$	Δf_{c-WB}

SCHEMATIC I: Parameter calculation table for each Block output

- From Schematic I, we can derive the values of N_1 & N_2

$$\Delta f_{WB} = N_2 N_1 \Delta f_{NB} \Rightarrow 72 \text{ kHz} = N_2 N_1 (18) \Rightarrow N_2 N_1 = 4000 \quad \text{I}$$

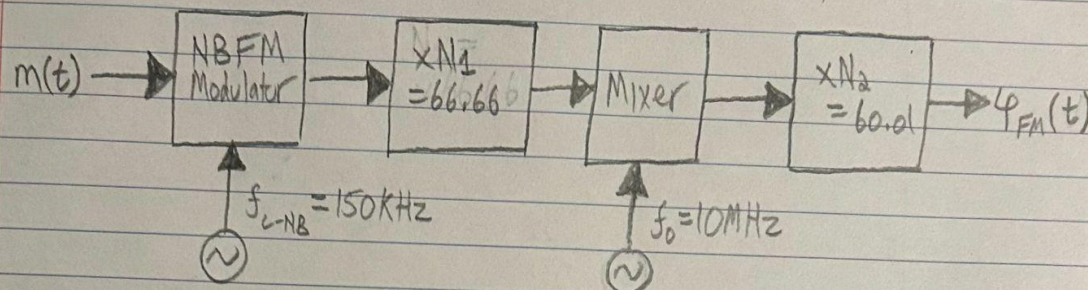
$$f_{c-WB} = N_2 \cdot f_o - N_2 N_1 \cdot f_{c-NB} \Rightarrow 100 \text{ kHz} = N_2 (10 \text{ MHz}) - N_2 N_1 (150 \text{ kHz}) \quad \text{II}$$

* Substituting equation I into II, we have:

$$\hookrightarrow 100 \text{ kHz} = N_2 (10 \text{ MHz}) - (4000) (150 \text{ kHz}) \Rightarrow N_2 = 60.01 \Rightarrow \frac{4000}{60.01} \Rightarrow N_1 = 66.66$$

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	NBFM	Multiplier-1	Mixer	Multiplier-2	WBFM
Δf	18 Hz	1.2 KHz	1.2 KHz	72 KHz	72 KHz
f_c	150 KHz	9.999 MHz	1 KHz	100 KHz	100 KHz

$\Delta f_{NB} = 18 \text{ Hz}$, $f_{c-NB} = 150 \text{ KHz}$ & $\Delta f_{WB} = 72 \text{ KHz}$, $f_{c-WB} = 100 \text{ KHz}$

The following calculations are computed for $N_1 = 66.66$ & $N_2 = 60.01$

$$(66.66)(18) = 1199.88 \text{ Hz}$$

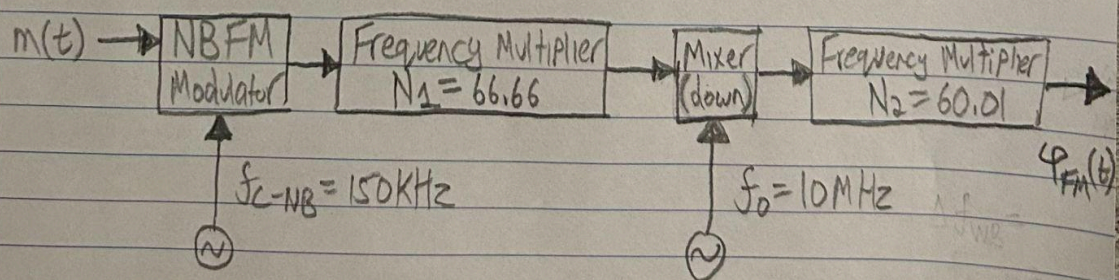
$$(66.66)(150\text{k}) = 9.999 \text{ MHz}$$

$$(10\text{M}) - (9.999\text{M}) = 1\text{KHz}$$

$$(60.01)(10\text{M}) - (60.01)(66.66)(150\text{k}) = 100\text{KHz}$$

* NOTE : Design targets are met

∴ the circuit that produces FM waves with design parameters for two (2) Stages of the multiplication factor using Armstrong's Indirect method is provided below for $N_1 = 66.66$ & $N_2 = 60.01$



CIRCUIT I : Final Block Diagram

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CONCLUSION

The robustness of Armstrong's Indirect Method for frequency modulation is effectively demonstrated by the simulation and design activities finished in this lab. We validated the spectral distribution predicted by Bessel functions in frequency modulation theory by using Simulink modeling to establish that centering the carrier at 25kHz provides for a clear visualization of the FM sidebands. A two-stage technique utilizing frequency multipliers and mixers to scale a moderate 18Hz deviation up to a conventional broadcast-quality 72kHz deviation was further demonstrated by the post-lab design exercise.

By integrating a 10MHz local oscillator with the initial 150kHz NBFM source, we illustrated how frequency translation can be used to reach the target 100kHz carrier without distorting the expanded frequency deviation. Ultimately, this assignment reinforces the principle that indirect FM generation effectively bypasses the stability limitations of direct frequency modulators, providing a practical engineering solution for high-fidelity communication transmitters.

REFERENCES

Zhao L., *ELE635: Communication Systems*. Toronto Metropolitan University, W2026. Accessed 2 April 2026.